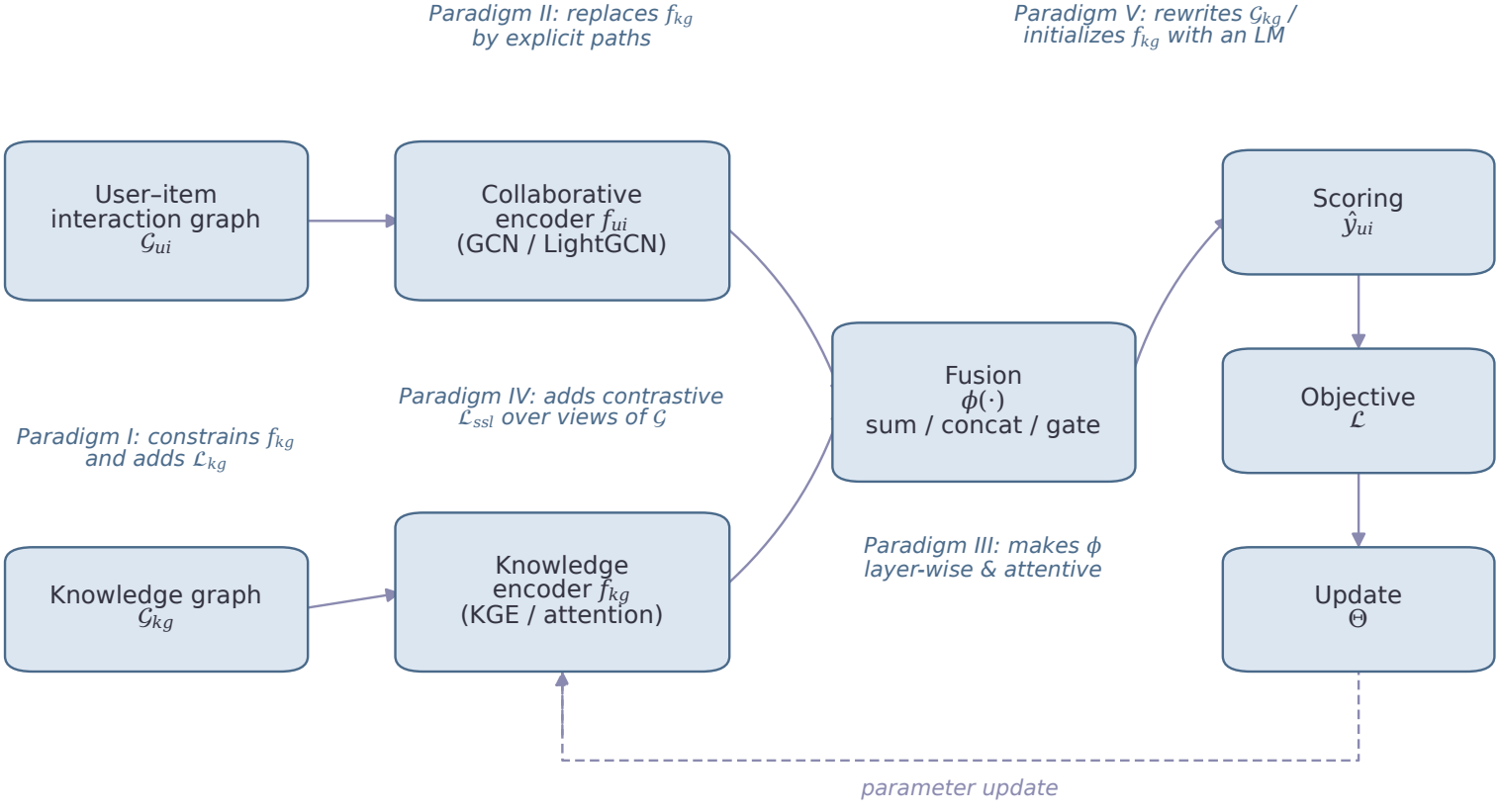
Paradigm II: replaces f_{kg} by explicit paths
Paradigm V: rewrites $\mathcal{G}_{kg}$ / initializes f_{kg} with an LM
User–item interaction graph $\mathcal{G}_{ui}$
Collaborative encoder f_{ui} (GCN / LightGCN)
Scoring $\hat{y}_{ui}$
Paradigm IV: adds contrastive $\mathcal{L}_{ssl}$ over views of $\mathcal{G}$
Fusion $\phi(\cdot)$ sum / concat / gate
Objective $\mathcal{L}$
Paradigm I: constrains f_{kg} and adds $\mathcal{L}_{kg}$
Knowledge graph $\mathcal{G}_{kg}$
Knowledge encoder f_{kg} (KGE / attention)
Paradigm III: makes ϕ layer-wise & attentive
Update Θ
parameter update